



**i 40-SS**



**i 40-PM**



**i 40-JB**



**i 40-DR**

**User Manual**



This guide is intended for i40 indicator users.  
It allows starting the equipment quickly. For electrical  
connection and installation, refer to the following manuals:

- Wiring diagram: 04-50-00 DD.
- Installation manual: 04-55-00 MI.

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## Operator setting

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## Operator configuration

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## Error messages

# Foreword

## Documentary conventions

### Pictograms



Shown on the device, they indicate a reference to the hereby manual for further information about the nature of potential dangers and all the measures that have to be taken to prevent them.



Major caution related to people's safety.



Note related to the preservation and correct maintenance of the device.



Note aimed at making the reading of the manual easier and/or at implementing the device optimally.

### Recycling

- In compliance with the requirements of the European Directive 2012/19/EU related to the electrical and electronic equipment waste, this device, including accessories and batteries must not be disposed of with regular household waste. This device must be disposed of according to local recommendations in a separate contained aimed at electrical and electronic equipment.

## ■ Environmental characteristics

- Temperature range:
  - in operation ..... -20°C / +55°C (+4°F / +119°F)
  - in charge with the battery option ..... -20°C / +30°C (+4°F / +94°F)
- Relative humidity without condensation ..... 10 % / 85 %h
- Minimum atmospheric pressure ..... 795 hPa (1)
- Protection index
  - i 40 SS and i 40 JB ..... IP66 (2) EN 60529 compliant
  - i 40 DR ..... IP30
  - i 40 PM ..... IP54 on front panel
- Standard power supply:
  - voltage ..... 9 V to 30 V DC (3)
  - typical max. consumption ..... 1.4 W
  - max. consumption option 4-20mA / 0-10 V ..... 1.8 W
  - max. consumption field bus option ..... 2.7 W
- Mains power supply:
  - voltage ..... 110 V / 230 V (4)
  - frequency ..... 50 Hz / 60 Hz
  - consumption ..... 0.35 A

## ■ Safety

The norm CEI 61010-1 indicates that in case of devices connected to an electrical plug, this plug must be installed near the device and easily accessible. The supply cord acts as cutting device. For the equipment permanently connected to the network, an easily accessible cutting device must be included to the fixed installation.

The mains tension must not fluctuate of more than 10% of its nominal tension.

The connection to the mains supply and the type of cord must comply with the safety regulations of the country in which the equipment is set up. If it is not the case, please contact your PRECIA-MOLEN agency.

1 Equivalent to a 2 000 m altitude.  
2 IP69 en option sur i 40 SS.  
3 22 V DC mini si option 4-20 mA.  
4 For Saudi Arabia only the 230 V is authorized.

## **Electromagnetic compatibility**

This equipment has been tested and declared compliant with the limits of the class A digital device limits, according to the European regulations, Section 15 of FCC rules (US Federal Communication Commission) and Canadian ICES-003 rules.

- These limits aim at offering sufficient protection against interferences when the device is used in a trade environment. If the device is used in a residential area, it may cause interferences and in these circumstances, the user will have to correct the situation at their own expenses.

Overview

Keyboard



| Function |                     |                                                                                                                                                                                          |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | Escape              | Escape a menu or the input without having it taken into account.                                                                                                                         |
|          | Menu                | <ul style="list-style-type: none"><li>- In operating mode, provides access to application setting.</li><li>- Allows entering configuration mode by pressing and holding power.</li></ul> |
|          | Validation          | Input in a menu or take into account the current input.                                                                                                                                  |
|          | Restart             | Hold these keys pressed to restart the indicator.                                                                                                                                        |
|          | Navigation keys     | Navigating in the menu or list options.<br>Indication < > in the help area.                                                                                                              |
|          | Zeroing             |                                                                                                                                                                                          |
|          | Taring              |                                                                                                                                                                                          |
|          | Gross weight recall |                                                                                                                                                                                          |



## ■ Navigation principle

- The navigation keys allow moving in menus and lists of predefined values. They are active when the symbols ‹ › are displayed in zone **3** of the screen (Voir "Screen", page 8.):



1. Scroll through the options at the same level (or possible values).



2. Validate the selected option (or the selected value).
- 

- In the same case for the consultation of recorded values:



Scroll through the displayed information.

---

- In case of navigation error:



Exit a menu, an item or option (move up one level).

---

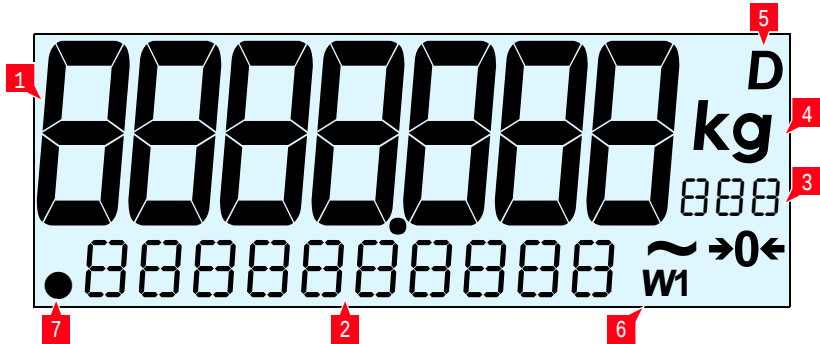
- Reach the store option (configuration mode):



Back to the configuration top level menu to reach the storage option.  
Actuate the button repeatedly, if necessary.

---

Screen



| Function       |                                                                                                                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title / Symbol | Operating mode                                                                                                                                                                                            |
| 1. Weight area | Displaying weight present on the receiver.                                                                                                                                                                |
| 2. User area   | Indications intended for the user.                                                                                                                                                                        |
| 3. Help area.  | <ul style="list-style-type: none"><li>- <b>NET</b> indication if tare taken into account.</li><li>- Additional unit.</li><li>- Indication of valid keys.</li></ul> See detail below.                      |
| 4. kg g t lb   | Unit.                                                                                                                                                                                                     |
| 5. D           | <ul style="list-style-type: none"><li>- No symbol. The displayed weight is a metrological reference.</li><li>- Symbol <b>D</b> displayed. The displayed weight is not a metrological reference.</li></ul> |
| 6. W1W2        | Range 1 or 2 is active.                                                                                                                                                                                   |
| ~              | The weight is unstable.                                                                                                                                                                                   |
| →0←            | The weight is in the zero area and stable.                                                                                                                                                                |
| 7. ●           | Confirmation of a key-press                                                                                                                                                                               |

## ■ Help area – Valid key indication detail

|       |                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------|
| < >   | Choice of menu, of value.                                                                                    |
| 0 - 9 | Numeric input.                                                                                               |
| 0 - Z | Alphanumeric input.                                                                                          |
| 1 - 6 | Numeric entry between two specified values (e.g.between 1 and 6).                                            |
| < >   | Indication of a menu (even if there is only one option).                                                     |
| ESC   | Escape a screen showing data display only.                                                                   |
| ↵     | Validation by  if necessary |

## ■ Basic parameters

### ■ Language

- **PARAMETERS / LANGUAGE**

Choose English or French.

### ■ Display

- **PARAMETERS / DISPLAY / ECO MODE** allows setting auto-extinction:

|     |                                                                                                                                                                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ON  | <ul style="list-style-type: none"> <li>- Turning off the backlight after 30s of inactivity, weight to zero.</li> <li>- After 1hour of inactivity, weight zero, the indicator turns off.</li> </ul> |
| OFF | Not auto-extinction.                                                                                                                                                                               |

- **PARAMETERS / DISPLAY / BRIGHTNESS** adjusts the backlight intensity: 3 levels or extinction.
- **PARAMETERS / DISPLAY / COLOR** sets the background colour of the screen: 7possible colours.

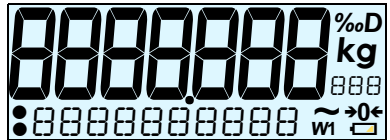
## Operator functions

### ■ Introduction

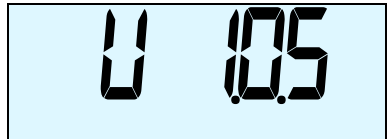
The operator functions are available when the indicator is powered on.

### ■ Power on

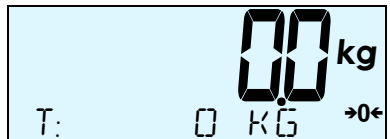
1. The instrument performs a self-test.  
Check the proper display operation (no defective segment).



2. The software version is displayed.



3. The self-test is complete, the instrument initialises the measurement and it is ready for use.



## ■ Zeroing

When the load receiver is empty and the displayed weight is not zero, the zero function allows to reset to zero the displayed weight (This function is limited to  $\pm 2\%$  of the maximum instrument range from the initial zero in trade use or to a configuration value outside legal metrology).

1. Receiver empty, the weight displayed is different from zero.



2. Select the reset to zero function (0↵).  
The displayed weight is set to zero.



## ■ Taring

The taring function allows storing the load weight in place on the receiver (container) as tare value and then obtain the product net weight (displayed weight = total weight - tare).

1. Place the container on the load receiver.



2. Select the taring function (0↵).  
The displayed weight corresponds to the net weight.



## ■ Initial zero setting

This zero setting can be made when the indicator is powered on. To do this, when powering on, the user presses the key 0↵ until the equipment starts.

This zero is used as reference to other zero devices.

## ■ Gross weight recall

1. The result of a net weighing is displayed.



2. Press the key . The screen changes colour (cyan) and the result is displayed as gross weight for a few seconds.



## ■ Range W2

In dual range, the measurement range selection is ensured automatically by the instrument based on the weight on the receiver. It is possible to force the instrument to use the range W2.

*Remark: This function is only available for dual range models.*

## ■ Information viewing

Press the keys  and  to scroll through the various information available.

1. Weight threshold value 1.



2. Weight threshold value 2.



3. Weight threshold value 3.



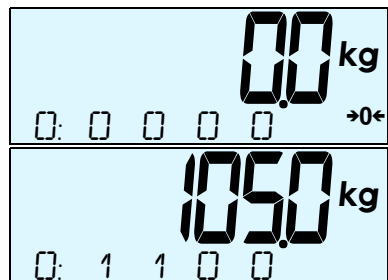
4. Weight threshold value 4.



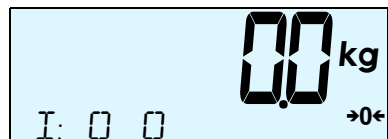
5. Tare value.



6. Output status.  
When one of the thresholds is exceeded, its status moves from 0 to 1.



7. Input status.

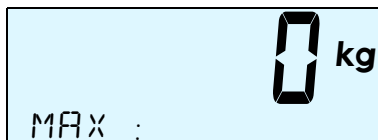


8. Weight in High Resolution.  
The last number corresponding to the high resolution is displayed in red.

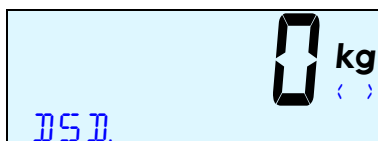


## **DSD reading**

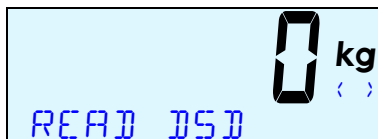
1. Press key  and scroll through the various functions with the navigation keys.



2. Select the function **DSD** and then validate ().



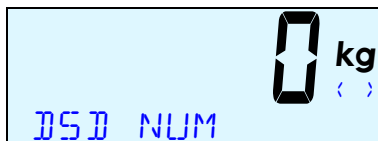
3. Select the function **READ DSD** and then validate (),



or select the function **DSD REF.** and then validate ().  
This reference allows identifying a bridge.

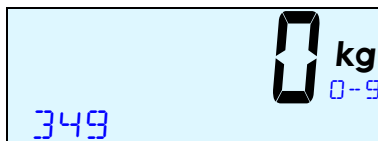


4. Select the function **DSD NUM** and then validate ().

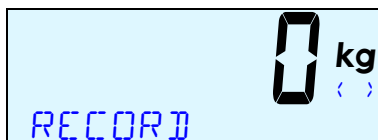


5. Enter the record number to be read using the keys  et  and then validate.

Remark: Enter 0 to display the last recording.



6. Validate once again to view the information corresponding to the record.





# Operator setting

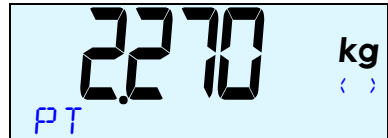
## Introduction

Press key  to access operator setting. Then use the keys  and  to access various available functions.

## Manually entering a tare value (preset tare)

This function allows to manually entering a tare value for the net weight display.

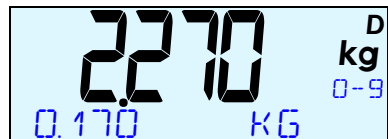
1. Select the preset tare function (PT) and then validate ().



2. The actual tare value is displayed.



3. Enter the new tare value by pressing the navigation keys:
  -  and  : Change the selected number value.
  -  and  : Move from one number to another.

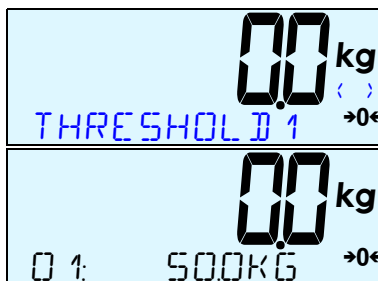


4. Validate this new value ().  
The value is taken as tare value. The weighing result is now displayed as Net.

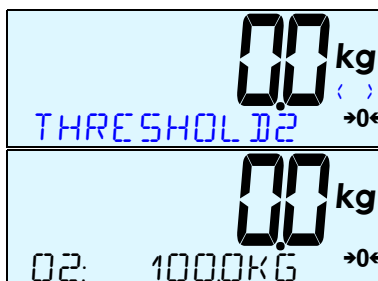


## ■ Threshold configuration

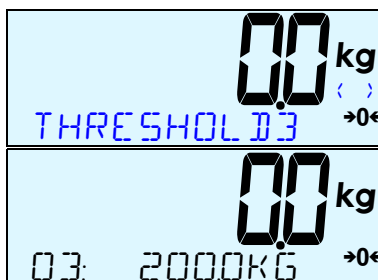
1. Select threshold 1 and then validate (↵).  
Set the threshold value and then validate to move to the next threshold.



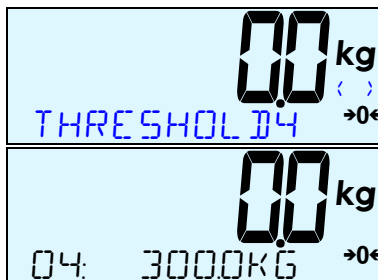
2. Select threshold 2 and then validate (↵).  
Set the threshold value and then validate to move to the next threshold.



3. Select threshold 3 and then validate (↵).  
Set the threshold value and then validate to move to the next threshold.

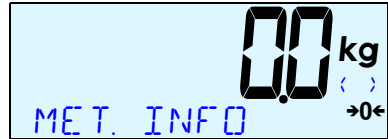


4. Select threshold 4 and then validate (↵).  
Set the threshold value and then validate to move to the next threshold.

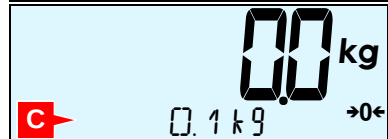
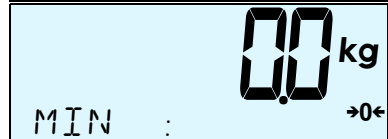
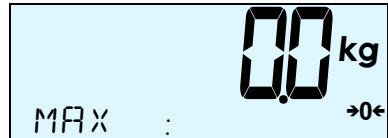


## ■ Metrological screen

1. Select the metrology label and then validate (↵).



2. Press key (F) to view the metrological data.  
The corresponding information is:
  - MAX: Equipment maximum range (A).
  - MIN: Equipment minimum range (A).
  - e: Step (C).



## Operator configuration

### Introduction

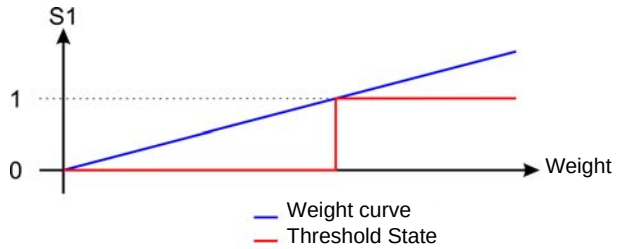
Press key  at indicator power on or restart to access the operator configuration.

### Threshold configuration

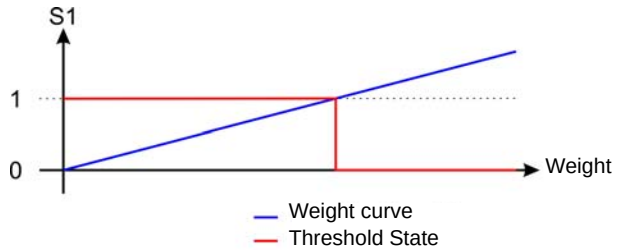
1. Select the function *THRESHOLD N* in the configuration menu.  
(**LOGIN / OPERATEUR / APP / THRESHOLD N**).
2. Enter the parameter *ACTIVE*.  
This parameter allows enabling the threshold:
  - **YES**  $\Rightarrow$  threshold enabled.
  - **NO**  $\Rightarrow$  threshold disabled.
3. Enter the parameter *WEIGHT*.  
This parameter allows selecting the threshold switching mode:
  - **NET**  $\Rightarrow$  the action is carried out on NET weight.
  - **GROSS**  $\Rightarrow$  the action is carried out on GROSS weight.
4. Enter the parameter *TRIGGER*.  
This parameter allows selecting the output switching time:
  - **>= THRESHOLD**  $\Rightarrow$  output enabled if the weight is greater than or equal to the threshold, otherwise disabled.
  - **<= THRESHOLD**  $\Rightarrow$  output enabled if the weight is less than or equal to the threshold, otherwise disabled.

## ■ Threshold switching

Threshold enabled at  
"1" state



Threshold enabled at  
"0" state



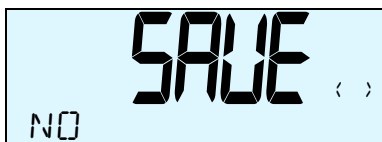
## Saving

When configuration parameters have been changed, the storing can be done in two ways:

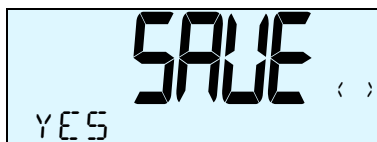
- going back to the highest level in the configuration menu:



The **SAVE** option screen is displayed for about 1s and then the choice for validating the modified parameters or not is offered.

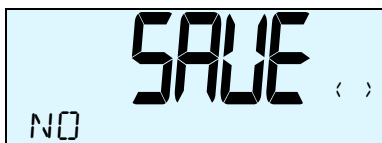


Select **NO** to return to the configuration menu without storing the changed parameters.



Select **YES** to return to the configuration menu after storing the changed parameters.

- by Pressing the keys  +  at the same time.  
Similarly, the **SAVE** option screen is displayed for about 1s, then the choice whether or not to confirm the modified parameters is proposed.



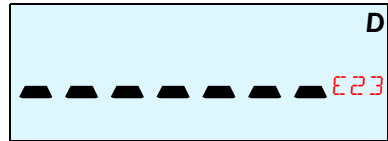
Select **NO** to turn the indicator off without storing the modified parameters.



Select **YES** to turn the indicator off after storing the modified parameters.

## Error messages

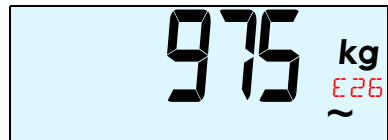
1. Instrument under-load.
  - Check the load receiver.
  - Carry out a reset to zero.



2. Instrument overload.
  - Remove part of the load.



3. Command denied
  - Example: Taring with an unstable weight



4. Zero rejected

The allowable weight limit for this function is exceeded.

  - Remove the load and press the zero key.

